

Tidy data

Lesson objectives

By the end of this session, you'll be able to:

- Describe what tidy data is
- Manipulate data into tidy formats
- Reshape data from wide to long format and vice-versa
- Use functions from the tidyverse to clean data of an example dataset
- Write reproducible code by applying concepts of literate programming

What is tidy data?

- Each variable has its own column
- Each observation has its own row
- Each value has its own cell

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174604898
China	1999	212258	127291272
China	2000	216766	128042583

variables

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174604898
China	1999	212258	127291272
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observations

country	year	cases	population
Afghanistan	99	745	19987071
Afghanistan	00	2666	20595360
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Brazil	00	80488	174604898
China	99	212258	127291272
China	00	216766	128042583

values

Example

country	1999	2000
Afghanistan	587	497
Brazil	93	88
China	3391	2034
...	...	...

What are the variables and observations in this data? Is this dataset tidy?

- One variable across multiple columns: not tidy!

country	1999	2000
Afghanistan	587	497
Brazil	93	88
China	3391	2034
...	...	...

- A tidy version:

country	year	cases
Afghanistan	1999	587
Afghanistan	2000	497
Brazil	1999	93
Brazil	2000	88
China	1999	3391
China	2000	2034
...	...	...

What about this dataset?

Student	Math test	English test	Essay
Beth	D	D	C
Carl	F	C	C
Erin	B	C	B
Joe	A	A	B

Is it in tidy format? If not, how would the tidy version look like?

A tidy version

Student	Assessment	Grade
Beth	Math test	D
Beth	English test	D
Beth	Essay	C
Carl	Math test	F
Carl	English test	C
Carl	Essay	C
Erin	Math test	B
Erin	English test	C
Erin	Essay	B
Joe	Math test	A
Joe	English test	A
Joe	Essay	B

Wide versus long

Student	Math test	English test	Essay
Beth	D	D	C
Carl	F	C	C
Erin	B	C	B
Joe	A	A	B

- Usually, long format is tidy data
- But it all depends on data and research questions!
- Good rule of thumb:
 - Variables are used to describe relationships
 - Different observations in the same variable to compare groups

Student	Assessment	Grade
Beth	Math test	D
Beth	English test	D
Beth	Essay	C
Carl	Math test	F
Carl	English test	C
Carl	Essay	C
Erin	Math test	B
Erin	English test	C
Erin	Essay	B
Joe	Math test	A
Joe	English test	A
Joe	Essay	B

Why tidy data?

- Keeps data structure consistent
- Easier to use lots of R functions

Let's practice!

Reflection

Now that you have finished data cleaning, look back to your data collection instrument. Is there any way that you could have set up the survey to make data cleaning simpler?

Type ideas in the chat.